

Erdős Problem #824

EP-824

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $h(x)$ count the number of integers $1 \leq a < b < x$ such that $(a, b) = 1$ and $\sigma(a) = \sigma(b)$, where σ is the sum of divisors function.

Is it true that $h(x) > x^{2-o(1)}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>